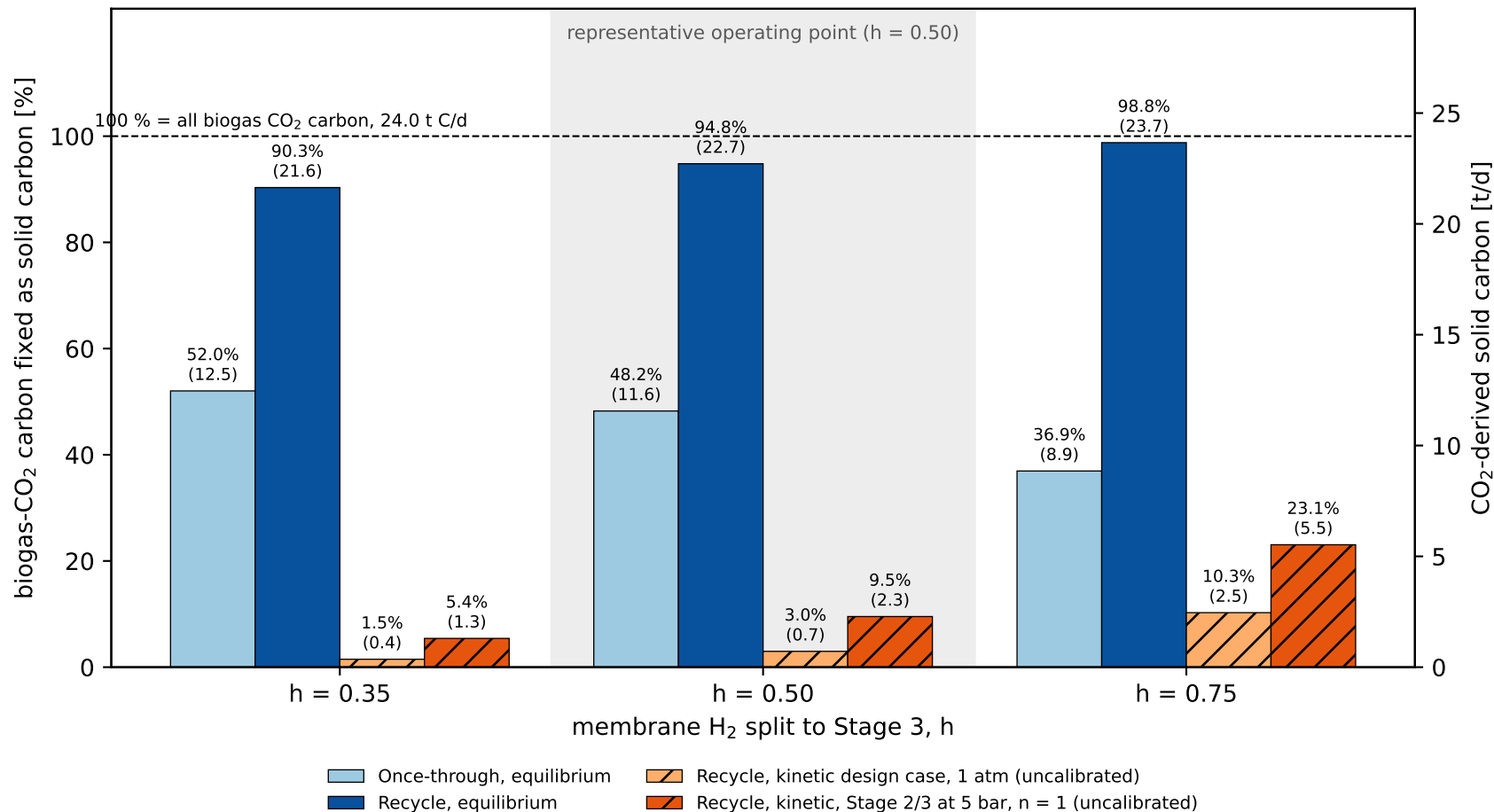


Fixation of biogas CO₂ as solid carbon, X4 configuration
(all biogas CO₂ to Stage 2, $y_{CH_4} = 0.60$, no DAC, no electrolysis; recycle $r_{CH_4} = 0.95$, purge 0.05)



Equilibrium: Gibbs minimisation at every stage. Kinetic: design case $\tau_1^* = 8.9$ s, Stage 2 two-reaction model ($\phi = 1$), Stage 3 reduced model at 650 K; rate constants uncalibrated. Bar labels: fixation % (CO₂-derived solid carbon, t C/d).